CELESTA

India High School Exoplanet Data Challenge

Official Dataset Guide | Celesta

NASA Kepler Objects of Interest (KOI) — Cumulative Table

Source: NASA Exoplanet Archive | DOI: 10.26133/NEA4 | Kepler Mission (2009–2018)

What Is This Dataset?

This dataset is the Cumulative Kepler Objects of Interest (KOI) table, maintained by the NASA Exoplanet Science Institute at IPAC, California Institute of Technology. It consolidates all vetted transit-like signals detected by NASA's Kepler Space Telescope over its primary mission lifetime, covering approximately 200,000 stars in the Cygnus constellation. Each row represents a single KOI — a periodic dimming event in stellar brightness that may or may not indicate a transiting exoplanet.

The dataset contains **9,564 rows** and **140 columns**, covering orbital parameters, stellar properties, transit geometry, and signal quality flags derived from the Kepler photometry pipeline.

Key Columns at a Glance

Column	Description
`koi_disposition`	Target variable: CONFIRMED, CANDIDATE, or FALSE POSITIVE
`koi_period`	Orbital period of the candidate planet [days]
`koi_duration`	Duration of the transit event [hours]
`koi_depth`	Fractional flux decrease during transit [ppm]
`koi_prad`	Planetary radius estimate [Earth radii]
`koi_teq`	Equilibrium temperature of the planet [K]
`koi_insol`	Insolation flux relative to Earth
`koi_impact`	Transit impact parameter

Target Variable — koi_disposition

Value	Meaning	Count (approx.)
CONFIRMED	Validated as a real planet by follow-up observations	~2,747
CANDIDATE	Transit-like signal not yet confirmed or ruled out	~1,978
FALSE POSITIVE	Signal explained by noise, eclipsing binary, or other artifact	~4,839

How to Use This Dataset

Your task: Build a machine learning classification model that predicts **koi_disposition** from the remaining columns — separating real exoplanet signals from false positives and unresolved candidates. This is the same problem NASA scientists tackle when vetting candidates from the Kepler pipeline.

1	**Explore**	Load the CSV, inspect distributions, check for class imbalance, and visualise key features.
2	**Clean**	Handle missing values and decide how to treat the CANDIDATE class (binary vs. three-class).
3	**Model**	Train a classifier of your choice (Random Forest, XGBoost, Logistic Regression, etc.).
4	**Evaluate**	Report Accuracy, Precision, Recall, F1-Score, and a Confusion Matrix.

5	**Explain**	Identify the most important features and explain what your model learned.

Note on koi_score: The column `koi_score` has been removed from this dataset. It is a pre-computed disposition score that would directly reveal the target variable, making the classification task trivial. The goal of this challenge is for you to build your own model from the raw transit and stellar features.

Original Source: https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=cumulative **DOI:** 10.26133/NEA4

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